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Consistent Portfolio Management: Alpha Construction

The Fundamental Law of Active Management tells us that good forecasts should directly translate to outperforming portfolios. Why, then, do we so often hear the frustrated lament that they do not? Can this discrepancy between the clear theoretical rigor and the negative practical experience be explained? For quantitative investors, the transfer coefficient introduced by Clarke et al. (2006) speaks to part of the discrepancy. But it is the inconsistent generation of a strategy, risk model, and expected returns that is largely accountable for those vexing gaps between theory and practice. In this paper we analyze the roots of this problem for quantitative management and propose a comprehensive approach that leads to the efficient implementation of quality signals into outperforming portfolios. We show that the discrepancy between theory and practice is partly explained by the incorrect interpretation of what we denote by "information", or IC, when applying the fundamental law. Additionally, we explain the gap between signal strength and realized performance by the incorrect implementation of the process that risk-adjusts the expectations of returns and uses the risk model as part of portfolio construction process. We provide a detailed analysis showing how each step of the signal and portfolio construction processes leads to the direct transfer of good forecasts into outperforming portfolios. Lastly, we illustrate through realistic examples how some common deviations from the consistent process we propose here impact this efficient transfer of information.

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Introduction

The main goal of mean-variance optimization is to transfer expectations of asset returns into a portfolio in an optimal, risk-adjusted manner. The Fundamental Law of Active Management describes how both the information in the expected returns and the transfer of this information into the portfolio are measured, and establishes the relationship between the two. If followed precisely, the Fundamental Law provides us with a theoretical framework that can be used to establish how this information translates into portfolio performance. However, it is not uncommon to see that portfolio managers are not able to translate good information (by their measure) into realized performance of their implemented portfolios. We believe that at the root of this problem is an unfortunate misunderstanding of how the Fundamental Law is applied.

Expected returns are a crucial component of the mean-variance optimization framework as is the asset covariance matrix and the portfolio construction strategy, represented by an objective and set of constraints. These three components are intimately connected in the quantitative investment process and yet they are all too frequently developed independently. In fact, some organizations are even structured such that one group develops an expected returns (alpha) model[1], a second group develops a factor risk model to represent the asset covariance matrix (or the risk model is procured from a third party), and a third develops and implements an investment strategy. As we will show in this paper, such a disconnected process can have unknown and undesirable consequences leading to a significant drag on performance.

When alpha modelers believe they have good forecasts and signals that perform well in practice, but do not see those returns realized by the implemented portfolio, they blame the inefficiencies of the market, and the portfolio management process for the lack of realized performance. Portfolio managers, on the other hand, understand that this loss of information is partly due to the imposition of constraints that are either mandated by the client or implemented as part of the investment strategy and partly due to other implementational frictions such as transaction costs. For this reason, there is recent research that has focused on understanding the role of constraints as well as measuring their impact via the transfer

[1] We do not distinguish between true alpha and exotic beta (exposure to portfolios with known and consistent risk premiums).

coefficient (see Clarke et al., 2006). But even the transfer coefficient may not explain the difference between expected and realized returns based on the performance of the expected returns. Frequently, neither alpha modelers nor portfolio managers pay much attention to the consistency of the process that is used to translate their good forecasts into optimal portfolios. In fact, we believe there is an anomaly that is created by the separation of these basic tasks that are part of the portfolio construction process and a lack of appreciation for the inconsistencies that are sometimes built into the quantitative process.

In this paper, we focus on alpha construction and we do so with consideration for how the alpha model is used within the overall investment process. When considering the portfolio construction strategy, we analyze how both the asset covariance matrix and some types of constraints affect the translation of alpha into the optimal portfolio. In particular, we consider constraints enforcing neutrality to risk factors as a part of the alpha construction process. We use these specific constraints as an example of how to construct a consistent investment process, but the same process can be extended to consider transaction costs, turnover restrictions, and other common considerations in the portfolio construction strategy.

While all of the frictions that are part of the portfolio construction process could be considered in the construction of expected returns, we do not propose doing so in this paper. (See Qian et al., 2007; Grinold, 2010, for methods of incorporating the IC horizon and transaction costs, respectively). Inclusion of all frictions could potentially be overly complex and impractical since different strategies typically share the same signal construction process. Furthermore, the research and implementation of such a process would be very strategy specific thus creating an alpha which is not portable across strategies. On the other hand, we typically see a lot of commonality between various strategies, so we can consider the commonalities in the creation of a consistent and portable process for the construction of expected returns.

When creating a portable alpha, we work under a framework where we construct and evaluate a target portfolio that represents at least some subset of the constraints that will be enforced in the strategies using that portable alpha. Therefore, our goal is to construct expected returns that will provide the target portfolio as the optimal solution to the investment strategy. In this paper, we only consider a set of constraints in our strategy such that it is possible to achieve the target portfolio.

Finally, we analyze the consequences of commonly encountered deviations from the consistent investment process proposed here. Sometimes these deviations are made unknowingly, while other times the deviations are implemented intentionally but without an understanding of the consequences. We provide analysis and numerical examples that illustrate how making seemingly innocuous alterations to our consistent investment process leads to a significant impact in the implemented portfolio. Specifically, we look at the effect of using inconsistent risk estimates in alpha construction and portfolio construction, the effect of not considering strategy constraints in the construction of the expected returns, and the effect of not appropriately risk-adjusting the expected returns for use in the portfolio construction strategy.

Background

This section provides the background needed to understand the consistent expected return (alpha) construction process that we discuss in the latter sections of this paper. For completeness, we try to succinctly describe the relevant aspects of prior work on consistent investment processes (see Black and Litterman, 1992; Grinold and Kahn, 2000; Sefton et al., 2004). We motivate and derive all results using our own point of view so as to better motivate the nuances that we believe are critical for the successful and practical implementation of this theory.

Motivation

Alpha construction generally begins with either an empirical observation of the behavior of asset returns such as low volatility stocks outperforming high volatility stocks or a financial principle such as firms with greater earnings per share providing a better return on investment. We refer to these drivers of return as factors throughout the paper. To test if these factors lead to outperformance in practice, and to precisely determine the extent of this outperformance, tests of signal strength are conducted.

One method that is frequently used is fractile analysis. Here, a long-short, dollar-neutral signal portfolio is constructed by holding a long position in the top fractile and an equal-sized short position in the bottom fractile. Such a signal portfolio whose returns represent a factor is known as a factor-mimicking portfolio (FMP). The return of the FMP is then analyzed and the quality of the signal is judged based on its return to risk ratio, or IR. The Fama-French factors are well-studied examples of such analysis (see Fama and French, 1992). Fractile analysis of many other factors are regularly reported in the literature by both academics and practitioners.

Signal portfolios constructed through fractile analysis may have undesirable exposure to other factors as well though, leading to a possible misrepresentation of the signal. For example, if the signal portfolio has exposure to countries, industries, or high-beta stocks, then the actual observed return of the portfolio may be due to these undesirable sources of return rather than the signal itself. Similarly, the volatility of the signal portfolio returns can be impacted by exposure to other sources of risk. Both scenarios may have a significant impact on the IR of a signal portfolio constructed from fractiles.

For this reason, Fama and French (1992) use a process, albeit somewhat simplistic, of neutralizing their SMB (small market capitalization minus big) and HML (high book-to-price minus low) factors to each other. Instead of simply using the top and bottom fractiles of a factor, the Fama-French factors take the top and bottom fractiles within each fractile of the other factor and combine them to create the neutralized signal portfolio. For example, to construct the SMB factor, the assets are sorted by market capitalization and split into two groups: small and big. The assets are also sorted on book-to-price and split into three groups: value, neutral, and growth. These two groupings are then combined to produce six intersecting value-weighted portfolios: small value, big growth, etc. Ultimately, the SMB portfolio is created by equal-weighting long positions in the three small portfolios and short positions in the three big portfolios. Therefore, the final SMB signal portfolio has no exposure to the neutralizing signal, book-to-price. While this process is intuitive, it is somewhat ad hoc and difficult to generalize to a larger group of factors. Nevertheless, the method is intuitive and we will refer back to such signal portfolios as *approximate FMPs* in the remainder of the paper.

In order to address the possibility of returns being explained by something other than the pure signal in a general multi-factor setting, we use the more rigorous construction of *pure* factor-mimicking portfolios. A pure FMP is a minimum variance portfolio that has zero exposure to all factors except the factor or signal of interest which we call the target factor. For now, we will assume that we have a single alpha signal. Positive exposure to this signal is expected to produce excess returns, or, in other terms, the signal is assumed to have information. We refer to factors for which the manager has no expectations of return as risk factors. Let X_R be a matrix of factor loadings to risk factors and X_A be a vector containing the raw alpha signal. Then the FMP construction problem can be written as

$$\begin{array}{ll} \text{minimize} & h^{\mathrm{T}} Q h \\ \text{subject to} & X_R^{\mathrm{T}} h = 0 \\ & X_A^{\mathrm{T}} h = 1, \end{array} \tag{1}$$

where h is the variable representing the optimal FMP and Q is a covariance matrix of asset returns. If $Q = X\Omega X^{\mathrm{T}} + \Delta^2$ is a factor risk model made up of the factors $X = \left[\begin{array}{cc} X_R & X_A \end{array}\right]$, then the objective can be equivalently replaced with $h^{\mathrm{T}}\Delta^2 h$.[2] In order to simplify our mathematical expressions and build intuition, we will consider the special case where $\Delta^2 = \sigma^2 I$, i.e., we will assume that the specific risks are homogeneous. In this case, the optimal solution to (1) is

$$\begin{aligned} h^* &= \left(I - X_R(X_R^{\mathrm{T}} X_R)^{-1} X_R^{\mathrm{T}}\right) X_A \left[X_A^{\mathrm{T}} \left(I - X_R \left(X_R^{\mathrm{T}} X_R\right)^{-1} X_R^{\mathrm{T}}\right) X_A\right]^{-1} \\ &= (I - P_R) X_A \left[X_A^{\mathrm{T}} (I - P_R) X_A\right]^{-1}, \end{aligned} \tag{2}$$

where $P_R = X_R(X_R^{\mathrm{T}} X_R)^{-1} X_R^{\mathrm{T}}$. [3]

Fundamental Law of Active Management

Armed with a time-series of pure FMPs representing our alpha signal, where each FMP is constructed using the exposures and asset covariance matrix available at the time, we can analyze the returns of these portfolios to gauge the quality of the signal. Let R_t be a vector of excess asset returns from time t to $t+1$. Then the return of the FMP in period t is computed as

$$\begin{aligned} h_t^{\mathrm{T}} R_t &= \left[X_A^{\mathrm{T}} (I - P_R) X_A\right]^{-1} X_A^{\mathrm{T}} (I - P_R) R_t \\ &= \left[X_A^{\mathrm{T}} (I - P_R)(I - P_R) X_A\right]^{-1} X_A^{\mathrm{T}} (I - P_R)(I - P_R) R_t \\ &= \left(\hat{X}_A^{\mathrm{T}} \hat{X}_A\right)^{-1} \hat{X}_A^{\mathrm{T}} r_t, \end{aligned} \tag{3}$$

[2]The objective can be equivalently replaced in the sense that both objectives $h^{\mathrm{T}} Q h$ and $h^{\mathrm{T}}\Delta^2 h$ will produce the same optimal solution. However, the actual optimal objective values will be different.

[3]The derivation of this result is given in the Technical Appendix.

where $\hat{X}_A = (I - P_R)X_A$ and $r_t = (I - P_R)R_t$.[4] The returns, r_t, are referred to as the residual returns. That is, they are the returns that remain after removing any portion of the excess asset returns that can be explained by the risk factors, X_R. The residual returns can, however, still be explained by the alpha factors.

With the assumption that $\Delta^2 = \sigma^2 I$, the portfolio and its return are intuitive. Equation (2) states that the FMP is simply the original signal, X_A, orthogonalized with respect to the risk factors, X_R, and then scaled to have unit exposure to X_A. If we assume that the $r_t \sim N(0, \sigma^2 I)$ which is consistent with the assumption that $\Delta^2 = \sigma^2 I$, then we can determine the information ratio (IR) as

$$
\begin{aligned}
IR &= \frac{h_t^{\mathrm{T}} R_t}{\sigma\sqrt{\left(\hat{X}_A^{\mathrm{T}}\hat{X}_A\right)^{-1}\hat{X}_A^{\mathrm{T}}\hat{X}_A\left(\hat{X}_A^{\mathrm{T}}\hat{X}_A\right)^{-1}}} \\
&= \frac{h_t^{\mathrm{T}} R_t}{\sigma\sqrt{\left(\hat{X}_A^{\mathrm{T}}\hat{X}_A\right)^{-1}}} \\
&= \frac{\left(\hat{X}_A^{\mathrm{T}}\hat{X}_A\right)^{-1}\hat{X}_A^{\mathrm{T}} r}{\sigma\sqrt{\left(\hat{X}_A^{\mathrm{T}}\hat{X}_A\right)^{-1}}} \cdot \frac{\sqrt{\sigma^{-2} r^{\mathrm{T}} r}}{\sqrt{\sigma^{-2} r^{\mathrm{T}} r}} \\
&= \frac{\hat{X}_A^{\mathrm{T}} r}{\sqrt{\hat{X}_A^{\mathrm{T}}\hat{X}_A}\sqrt{r^{\mathrm{T}} r}}\sqrt{\sigma^{-2} r^{\mathrm{T}} r} \\
\Rightarrow E[IR] &= IC \cdot \sqrt{N},
\end{aligned}
$$

where N is the number of assets in the investment universe. Thus, we arrive at the Fundamental Law of Active Management.[5]

The Fundamental Law is frequently misinterpreted due to the IC being computed in ways that do not directly lead to improvements in IR. In the analysis above, we derived the correct IC, which is a particular relationship between *residual returns* and a *pure signal*. We refer to a pure signal as a pure FMP that represents an alpha signal. Let us examine what happens when either residual returns or pure signals are not used in the computation of the IC. First, consider the effect of not orthogonalizing the signal to the risk factors, that is, consider the computation of the IC when using the residual returns and X_A rather than $\hat{X}_A$. Suppose that the market beta is one of the risk factors and let each asset's exposure to the market be represented by β. Suppose that our raw signal has exposure to this risk factor, i.e., $\beta^{\mathrm{T}} X_A \neq 0$. Since the portion of returns explainable by β have been removed from r, the strength of the signal as given by the IC when it is computed using the raw signal will generally underestimate the actual strength of the pure signal. In this case, the numerator of the IC would be unchanged, but the denominator would change because $\|\hat{X}_A\| \leq \|X_A\|$. If, in addition to using X_A instead of $\hat{X}_A$, we also use R rather than r, the IC value becomes

[4] Note that P_R is a symmetric projection matrix and therefore $P_R = P_R^2$ and $(I - P_R) = (I - P_R)^2$.

[5] IC is sometimes defined to be the correlation between r and $\hat{X}_A$, but this is only true if both r and $\hat{X}_A$ have mean zero.

even more misleading. To see this, we use the fact that $\hat{X}_A$ is the residual vector to the least-squares solution of the linear model

$$X_A = X_R u + \hat{X}_A \tag{4}$$

and the residual returns, r, are the residuals to the least-squares solution of

$$R = X_R f_R + r, \tag{5}$$

to represent the total excess return of the raw signal as

$$\begin{aligned} X_A^{\mathrm{T}} R &= (\hat{X}_A + X_R u)^{\mathrm{T}} (X_R f_R + r) \\ &= \hat{X}_A^{\mathrm{T}} r + u^{\mathrm{T}} X_R^{\mathrm{T}} X_R f_R. \end{aligned}$$

So, the return that is used in the correct computation of the IC, $\hat{X}_A^{\mathrm{T}} r$ is actually skewed by the value $u^{\mathrm{T}} X_R^{\mathrm{T}} X_R f_R$ which is an undeterminable value that may be a significant and lead to a misinterpretation of IC.

In summary, when computed correctly, the IC provides us with the strength of the pure signal. The Fundamental Law tells us how the strength of this signal leads directly to the performance of the portfolio – if the signal is exactly represented in the portfolio. In order for this to occur, the ultimate portfolio should be proportional to the FMP, h^*. And if this holds, then the optimal portfolio is exposed to the raw alpha signal, but not to any of the risk factors. This is what we will strive for so that our final portfolio will reflect our IC calculated using residual returns and our pure signal. We say that a signal is *ideally represented* if the optimal portfolio is a scalar multiple of the target portfolio, h^*.

Computing Expected Returns

Armed with a signal portfolio, we now turn our attention to the construction of expected returns. When we measured the IC, we actually considered the return of a target portfolio. If we want to get the same return in our final optimized portfolio, then we need to consider the effect of using our expected returns in a mean-variance optimization framework. We might expect that we only need to get maximal risk-adjusted exposure to the signal or signal portfolio. However, such a process does not provide us with our goal of achieving a portfolio that is proportional to our target.

Consider the following portfolio construction problem where we maximize risk-adjusted exposure to our expected returns, α:

$$\max \alpha^{\mathrm{T}} h - \lambda h^{\mathrm{T}} Q h. \tag{6}$$

The optimal solution to this problem is $(Q^{-1}\alpha)/(2\lambda)$. For a general covariance matrix Q, the solution to (6) will have exposure to risk factors and therefore cannot be proportional to the target, h^*, since h^* does not include such exposures.

To eliminate this undesirable exposure, we can constrain the optimization to have zero exposure to the risk factors as we did with our original FMP construction. That is, we can extend the problem as

$$\begin{array}{ll} \text{maximize} & \alpha^{\mathrm{T}} h - \lambda h^{\mathrm{T}} Q h \\ \text{subject to} & X_R^{\mathrm{T}} h = 0. \end{array} \tag{7}$$

This eliminates exposure to the risk factors. However, the solution to (7) may still not be proportional to our target. The reason for this is that Q may risk-adjust α differently than did the alpha construction process. For example, suppose that our alpha is only partially spanned by the factors in Q. Then the orthogonal portion of α will be over-weighted relative to the spanned portion causing the optimal solution of (7) to not be proportional to h^*. For a general covariance Q, the solution to (7) is

$$h = \frac{1}{2\lambda} Q^{-1/2} \left(I - Q^{-1/2} X_R (X_R^{\mathrm{T}} Q^{-1} X_R)^{-1} X_R^{\mathrm{T}} Q^{-1/2} \right) Q^{-1/2} \alpha. \tag{8}$$

If we let $\alpha = Qh^*$, then the solution to both (6) and (7) is a multiple of h^*. [3] This value of α is the implied alpha of the target portfolio, h^*. As long as the same risk model used to compute the implied alpha is also used in the portfolio construction problem, then the optimal solution to the unconstrained mean-variance optimization will be our target portfolio, h^*. The choice of α may seem counter-intuitive. Even though the target portfolio has no exposure to the risk factors, the alpha constructed using the implied alpha of the target portfolio may have exposure to risk factors. Thus, to get back the desired portfolio, the manager must effectively include expectations of the risk factors into the expected returns – at least for the unconstrained portfolio construction problem.

If we let $\alpha = \hat{X}_A$ or $\alpha = X_A$, then the solution of (6) may have exposure to the risk factors, but the solution of (7) is actually a multiple of h^*.[6] So, if we use a single alpha signal that is orthogonalized to the risk factors as our expected returns in a portfolio construction strategy that constrains the exposure to each risk factor be zero, then we get the target portfolio using expected returns that do not have exposure to the risk factors. In the next section, we will show how the weights on the raw or orthogonalized alpha signals must be determined in order to get the same result in the multiple alpha signal case.

To this point, we have reviewed the meaning of IC and showed that it is a measure of the strength of an alpha signal that can be directly translated into performance of the portfolio if and only if the alpha signal is ideally represented in the portfolio. Then, we showed that in order to get the FMP represented in the portfolio, our expected returns should be set to the implied alpha of the target portfolio in the unconstrained portfolio construction strategy or to the raw or orthogonal alpha signals so long as risk factor neutrality constraints are present in the portfolio construction strategy. In the absence of any other constraints, the optimal solution to the mean-variance portfolio problem will be the target portfolio if this recipe is followed.

We have described a consistent investment process in a long-short setting[7]. Both the pure FMP and optimal portfolio are not constrained to be long-only. In a long-only setting where the manager's goal is to outperform a benchmark, the consistent investment process we have

[6] Here, we refer to the orthogonalized form $\hat{X}_A$ as an othogonalized version of the raw signals that is not risk adjusted. We first defined $\hat{X}_A$ to be a scaled version of the target portfolio, but it was also the portion of the raw alpha signals that were orthogonal to the risk factors due to the assumption that $\Delta^2 = \sigma^2 I$. In the more general setting we let $\hat{X}_A$ be the orthogonal part of X_A and not h^*. Using h^* as expected returns is an inconsistent process because doing so would lead to the alpha signals being risk adjusted twice, i.e., the optimal portfolio would be proportional to $Q^{-1}Q^{-1}\alpha$.

[7] If a market(beta) factor is present in the set of risk factors, then the long-short portfolio will also be dollar(beta) neutral.

discussed is still applicable. In the relative performance setting, the target portfolio can be thought of as the active portfolio. A long-only constraint may prevent a manager's active portfolio from fully representing the target portfolio, but we will focus on the construction of expected returns and will continue to focus on the unconstrained and risk-factor neutral cases in this paper. The impact of such implementational constraints is beyond the scope of this paper.

Multiple Alpha Signals

Now that we have seen the simplistic case where we worked with a single alpha signal, we consider a generalization of the methodology where multiple alpha signals are present. There are a a few differences in the case of multiple signals that we must consider.

Following the steps of the single factor case, we return to the Fundamental Law and consider a target portfolio. We will let this target portfolio be a combination of the pure signal portfolios corresponding to each of our alpha signals. Since the target portfolio will be a combination of pure signal portfolios, we will have to construct these portfolios and gather certain statistics that will be used in the construction of the target portfolio. First, we must determine both the quality of each signal and the correlation of returns of the alpha-generating FMPs. Second, we must construct a target portfolio using the signal strength and correlation information of the multiple signals. The third and final step of computing the expected returns works exactly as it did in the single alpha-signal scenario; the expected returns are set to the implied alpha of the target portfolio.

Now, consider the first step of determining the information in each alpha signal. In the single alpha signal scenario, we constructed a pure FMP that was analyzed to determine the amount of information in the signal. In the multiple-signal setting, we can follow the same process. We must compute the FMP of each alpha signal and to ensure that the FMP is pure, we need to purify the FMP with respect to both the risk factorsand the other alpha signals. The FMP construction problem can then be written as

$$\begin{array}{ll} \text{minimize} & h^{\mathrm{T}} Q h \\ \text{subject to} & X_R^{\mathrm{T}} h = 0 \\ & X_A^{\mathrm{T}} h = e_j, \end{array} \tag{9}$$

where X_A is a matrix whose columns are the raw alpha signals and e_j is the jth column of an identity matrix of size k where k is the number of alpha signals. Let h_j be the FMP associated with alpha signal j. As in the single signal case, the return of the FMP is simply $h_j^{\mathrm{T}} R$.

The next step is to combine the FMPs to create a target portfolio in a multi-factor setting. This will create a target portfolio having exposure to each of the alpha signals and to none of the risk factors. Now, we consider how we will weight each of these alpha signal portfolios in the target portfolio. We will follow our earlier arguments regarding the Fundamental Law and choose weights so as to maximize IR (equivalently IC). Let ω be a vector of weights on the FMPs to be determined such that our target portfolio will be $\sum_{j=1}^{k} h_j \omega_j$. Let $E[f_A]$ be the expected return of the FMPs and Ω_A be the covariance matrix of the returns of the

FMPs. Then, to determine the optimal weights on the FMPs, we consider the problem[8]

$$\text{maximize } E[f_A]^{\mathrm{T}}\omega - \lambda\omega^{\mathrm{T}}\Omega_A\omega. \tag{10}$$

The optimal solution to (10) provides the weights that are applied to the FMPs in determining the target portfolio. In the case where h_j is computed assuming $Q = \sigma^2 I$, the final target portfolio can then be represented as [3]

$$\sum_{j=1}^{k} h_j\omega_j = (I - P_R)X_A[X_A^{\mathrm{T}}(I - P_R)X_A]^{-1}\omega. \tag{11}$$

The target portfolio will be neutral to all risk factors since each of the individual FMPs is also neutral to the risk factors. If we then take the implied alpha of this portfolio, we get the final expected return vector

$$\alpha = Q(I - P_R)X_A[X_A^{\mathrm{T}}(I - P_R)X_A]^{-1}\omega. \tag{12}$$

In practice, we do not need to compute the FMP corresponding to each individual alpha signal in order to compute the target portfolio. Instead, the target portfolio can be determined by solving the problem [3] :

$$\begin{array}{ll} \text{minimize} & h^{\mathrm{T}}Qh \\ \text{subject to} & X_R^{\mathrm{T}}h = 0 \\ & X_A^{\mathrm{T}}h = \omega. \end{array} \tag{13}$$

Furthermore, we do not need to compute the FMP of each alpha signal for the purposes of determining $E[f_A]$ or Ω_A. We can alternatively compute the FMP returns by considering a different grouping of the terms in (3). That is, rather than directly compute an FMP and look at the return of the portfolio, we can compute the returns of the FMP directly by considering the following model of asset returns:

$$R = X_R f_R + X_A f_A + \varepsilon. \tag{14}$$

The least-squares solution of this model satisfies the linear system

$$X_A^{\mathrm{T}}(I - P_R)X_A f_A = X_A^{\mathrm{T}}(I - P_R)R \tag{15}$$

$$X_R^{\mathrm{T}}X_R f_R = X_R^{\mathrm{T}}(R - X_A f_A). \tag{16}$$

Here, we can see the equivalence between f_A defined in (15) and $h^{\mathrm{T}}R$ as defined in (3). That is, the linear regression gives us the returns of the FMPs. Estimates of the expected returns and covariances of the FMPs, i.e, $E[f_A]$ and $\mathrm{cov}[f_A]$, can then be computed for use in the construction of a target portfolio. Thus, we actually get the desired factor returns needed for alpha construction for "free" by considering a returns model that has both risk factors and alpha factors. Later, we show how the use of a factor risk model that contains the alpha factors in the construction of the target portfolio can have a significant impact on the overall investment process.

[8] Any practical constraints such as $\omega \geq 0$ or $\sum_j \omega_j = 1$ can be added if desired and still remain a consistent investment process.

We refer to the method we have outlined here as the Grinold and Kahn (GK) approach as the steps that we have described here are mostly contained in Grinold and Kahn (2000). It is a consistent investment process, but it is not the only one. The Black-Litterman model is an equivalent approach to alpha construction in a certain sense and can be thought of as a generalization of the method we just described. We consider the "active portfolio management" version of the Black-Litterman model where the equilibrium expected active returns are zero (see Da Silva et al., 2009). Let the investor views be given as $P\alpha = q$ and known with certainty.[9] Under this scenario, $\alpha = QP^{\mathrm{T}}[PQP^{\mathrm{T}}]^{-1}q$. In the GK approach, our views are on the factor-mimicking portfolios. If we consider the same views in the Black-Litterman approach, the two methods are equivalent if $PQP^{\mathrm{T}} = \Omega_A$ and $q = E[f_A]$. To see this, first note that PQP^{T} is the covariance matrix of the views – factor mimicking portfolios in this case. Next, note that $\omega = [PQP^{\mathrm{T}}]^{-1}q$ is the optimal solution to the mean-variance problem

$$\max q^{\mathrm{T}}\omega - \frac{1}{2}\,\omega^{\mathrm{T}}PQP^{\mathrm{T}}\omega. \tag{17}$$

Since Ω_A is the variance of the returns of the FMPs, we have $\Omega_A = PQP^{\mathrm{T}}$. So, the Black-Litterman model is also a consistent investment process. Note that it implicitly takes the implied alpha of a target portfolio as the ultimate expected returns as does the GK approach.

While both the GK and BL approaches take the implied alpha of a target portfolio, it is very common to see managers constructing expected returns as just a linear combination of the raw alpha signals. It is possible to created expected returns in this way such that the target portfolio is ideally represented. Let $\hat{X}_A = (I - P_R)X_A$ where X_A is a matrix whose columns are the raw alpha signals. Then, if we let $\alpha = \hat{X}_A\gamma$ for some vector of raw alpha signal weights γ, we can write the optimal solution of (7) as

$$h = \frac{1}{2\lambda}Q^{-1/2}\left(I - Q^{-1/2}X_R(X_R^{\mathrm{T}}Q^{-1}X_R)^{-1}X_R^{\mathrm{T}}Q^{-1/2}\right)Q^{-1/2}X_A\gamma. \tag{18}$$

So, if we let

$$\gamma = \left[X_A^{\mathrm{T}}Q^{-1/2}(I - P_R^Q)Q^{-1/2}X_A\right]^{-1}\omega, \tag{19}$$

then the solution to (7) is proportional to h^* when $\alpha = \hat{X}_A\gamma$. [3] The same weights applied to X_A directly also gives the target portfolio. This value of γ is equal to the covariances between the target portfolio and the FMPs corresponding to the alpha factors. That is, $\gamma_j = h_j^{\mathrm{T}}Qh^*$. [3] Note, however, that using $\alpha = X_A\gamma = \hat{X}_A\gamma$ as the expected returns does not provide a consistent process. This value of α only provides the target portfolio as the optimal solution when the exposures to risk factors are constrained to be zero in the portfolio construction strategy. Such a portfolio construction problem will generally have a transfer coefficient less than one. We say that a process is perfectly consistent if and only if it obtains a multiple of the target portfolio with a transfer coefficient equal to one. The reason for this is that it should be a goal to achieve a transfer coefficient of one so that the expected returns are completely transferred to the portfolio. If the transfer coefficient needed to obtain the target portfolio is less than one, say 0.5, then it is virtually impossible for the portfolio manager

[9]The general form of the Black-Litterman model allows for uncertainty to be placed on each view.

to manage the portfolio construction part of the process. Improving the transfer coefficient could actually take the ultimate portfolio further away from the target portfolio.

Constructing expected returns is not only about choosing the right signals and creating good FMPs. It is also about forecasting the returns of the FMPs. The factor return forecasts, $E[f_A]$, play a critical role in determining ω. Recall that the target portfolio is a linear combination of the individual FMPs, weighted by ω. Therefore, we can write the IC of the target portfolio as[10]

$$\begin{aligned} IC &= \frac{\omega^{\mathrm{T}} f_A}{\sqrt{\omega^{\mathrm{T}}\Omega_A\omega}\sqrt{f_A^{\mathrm{T}}\Omega^{-1}f_A}} \\ &= \frac{E[f_A]^{\mathrm{T}}\Omega^{-1}f_A}{\sqrt{E[f_A]^{\mathrm{T}}\Omega^{-1}E[f_A]}\sqrt{f_A^{\mathrm{T}}\Omega^{-1}f_A}}. \end{aligned} \tag{20}$$

There are a couple of interesting points about (20). First, we can see that factor timing comes into play here because the IC is essentially the correlation between risk-adjusted factor returns and risk-adjusted expectations of the factor returns. Therefore, the performance of the target portfolio depends precisely on how well the factor returns can be estimated. Forecasting of factor returns is difficult and it is thus common practice to search for factors that have consistent returns. Second, we see that the risk estimation of the factor returns plays a crucial role in determining the IC and thus the IR of the ultimate portfolio. We have thus far taken it for granted that the covariance estimates used to construct the target portfolio is accurate thus leading to an optimal risk-adjusted weighting of the signal portfolios, but this may not be the case in practice.

Interaction of Alpha Signals and Risk Models in Portfolio Construction

The risk model impacts the portfolio construction process along multiple dimensions. First, the risk model affects the computation of the weights on the signal portfolios that make up the target portfolio. Second, the risk model affects the computation of the implied alpha of the target portfolio. If our process is consistent and we use the same risk model in both computing the implied alpha that define the expected returns and as the asset covariance matrix in the portfolio construction problem, then the effects from the two offset each other, guaranteeing that we obtain the target portfolio in the absence of any other constraints that reduce the transfer coefficient. Lastly, the risk model affects the construction of the optimal portfolio when constraints do prevent the optimal solution from being a multiple of the target portfolio. This is an involved topic that is beyond the scope of this paper, but we do make one important point here before moving to our main interest of investigating the impact on the FMP weights in constructing the target portfolio.

[10] This formulation of the IC assumes that the risk estimates of the factor mimicking portfolios given by Ω_A are consistent with those given by Q. This is actually not the case if Ω_A is a block of the factor covariance matrix that makes up Q. The reason is that the risk as estimated by Q will contain not only the factor portion of risk which will be equivalent to that estimated by Ω_A, but it will additionally have the specific risk portion of the FMPs. Therefore Q always overestimates the risk of the FMPs used to construct Q.

Suppose that the portfolio manager's mandate is to be neutral to a specific set of factors: dollar-neutral, beta-neutral, industry-neutral, and country-neutral. That is, the portfolio manager wants zero active exposure to each of the factors in these categories. Then the portfolio manager should consider each of these factors among the risk factors when constructing both the risk model and the expected returns. That is, the set of risk factors should be included in X_R. If neutrality constraints are applied to any type of factor that is not included in X_R, then the constraint can potentially affect the return of the target portfolio, or "eat alpha". If these risk factors are not used as part of the construction of expected returns, the information in the alpha signals is not transferred to the optimal portfolio. This loss of information is measured by the transfer coefficient and is a part of an extension to the Fundamental Law of Active Management introduced by Clarke et al. (2006). The transfer coefficient is intended to measure implementation effects such as turnover constraints, long-only restrictions, etc. There is certainly no need to suffer an additional drag in the performance of the target portfolio in the absence of such implementational constraints. The case of not constraining a factor that is in X_R is not a concern to us here if we let $\alpha = Qh^*$. The target portfolio does not contain exposure to X_R anyway, so not constraining these factors has no impact in the absence of other constraints.

Let us now analyze the effect of the risk model on the weights on the FMPs representing the alpha signals. In the GK(BL) approach, the weights on the individual FMPs(views) are determined based on the expected returns and covariances of the FMPs(views). Rather than assuming a custom returns model is used to construct Ω_A as we outlined in this paper, suppose that the factor mimicking portfolios are generated using (9) with a third party risk model, Q. The risk estimates of the FMPs are taken to be the risk estimates provided by Q. That is, $(\Omega_A)_{ij} = h_i^{\top} Q h_j$. Suppose estimates of $E[f_A]$ are provided by the portfolio manager and (10) is optimized to compute ω. Problem (13) is then solved to compute the target portfolio. These steps are implicitly those taken when using the Black-Litterman model.

If the systematic risk associated with one of the alpha factors is not captured by Q, then the variance of the factor and covariance with all other factors will be misestimated by Ω_A. Specifically, the risk of the alpha factor will be underestimated. This will lead to the signal being over represented in the target portfolio. Since all subsequent steps of a consistent investment process are designed to construct a portfolio that is as close to the target as possible, the final portfolio will also overweight this particular alpha signal.

This misestimation of risk of the FMP thus leads to an underestimation of risk of the final portfolio. Even worse, the misestimation of risk leads to a non-optimal allocation of pure alpha signals in the final portfolio thus leading to a reduction in realized IR.

Implications of Deviating from Consistent Process

We have described a consistent process for constructing expected returns to be used in mean-variance optimization. It is common for portfolio managers to follow at least some steps of a consistent process, yet we see few managers who follow an entirely consistent process. In this section, we will look at some common deviations from the consistent process that we described and investigate the implications of such deviations. We first introduce a statistic that we will use to measure the extent of the deviation.

Clarke et al. (2006) introduced the Transfer Coefficient (TC) to measure how far the ultimate portfolio is from a certain target portfolio. This target portfolio is the unconstrained optimal portfolio. If we follow their extension to the Fundamental Law of Active Management that states

$$IR = IC \cdot TC \cdot \sqrt{N} \tag{21}$$

where $-1 \leq TC \leq 1$, then the information used to compute the IC will be fully transferred to the final portfolio if and only if the TC is equal to one. Mathematically, the TC is defined as (see Clarke et al., 2006):

$$\mathrm{TC} = \frac{\alpha^{\mathrm{T}} w}{\sqrt{\alpha^{\mathrm{T}} Q^{-1} \alpha}\sqrt{w^{\mathrm{T}} Q w}}, \tag{22}$$

where Q is the asset-asset covariance matrix and w is the vector of portfolio weights. Clarke et al. (2006) discuss how the TC can be viewed as the "correlation" between the risk-adjusted expected returns and the risk-weighted portfolio, i.e., between $Q^{-1/2}\alpha$ and $Q^{1/2}w$. We can also think of the TC as the "correlation" between the risk-weighted ultimate portfolio and risk-weighted unconstrained optimal portfolio. The TC is a measure of how far the optimal portfolio is from the unconstrained optimal. Its value can only deviate from one in the presence of constraints or other implementational frictions such as transactions costs that drive the optimal portfolio away from the unconstrained optimum.

Since we define $\alpha = Qh^*$ in our ideal scenario, we can write the transfer coefficient as

$$\mathrm{TC}^* = \frac{w^{\mathrm{T}} Q h^*}{\sqrt{(h^*)^{\mathrm{T}} Q (h^*)}\sqrt{w^{\mathrm{T}} Q w}}. \tag{23}$$

Here, Q should be the covariance matrix used in the ultimate portfolio construction problem. We define this alternative form of the transfer coefficient as TC^* because it is only equivalent to TC when we define $\alpha = Qh^*$. When α is constructed in a different manner, TC^* is the "correlation" between the risk-weighted ultimate portfolio and risk-weighted target portfolio, i.e., $Q^{1/2}h^*$ and $Q^{1/2}w$. While the TC defines how far the optimal portfolio is from the unconstrained optimal portfolio, TC^* defines how far the optimal portfolio is from the target portfolio. And unlike the TC that can only be impacted by constraints, TC^* is a holistic measure that is also impacted by the alpha construction process and the risk model used in the portfolio construction strategy.

Equation (23) has important implications when considering the Fundamental Law in practice. We do not automatically assume that the investment process is a consistent one where (22) and (23) are equivalent. The Fundamental Law given in (21) suggests that the return of the IC will be realized if the TC as determined by (22) is equal to one. However, this is only true if the IC is measured in a way that is consistent with the theory used to derive (21).

To see how different the two may be in even a reasonable scenario (as opposed to an extreme pedagogical scenario), consider the following example. Let X_{mom}, X_{roe}, and X_{stp} be raw alpha signals corresponding to momentum, return-on-equity, and sales-to-price, respectively. We generate a time-series of factor returns for these three alpha signals as well as for a set of risk factors over a global universe of about 10000 global equities by computing a weighted least-squares estimate of (14) in each period. Using these factor returns, we compute a covariance matrix, Ω_A, of the factor returns corresponding to the three alpha factors.

We let the expected returns of each of the factors be one and solve a constrained version of (10) to compute a set of optimal alpha factor weights, ω, such that $\omega \geq 0$ and the elements of ω sum to one. We then solve (13) to determine our target portfolio, h^*.

In order to illustrate how TC and TC^* are impacted by the expected returns, the risk model, and the strategy, we consider three different expected returns, two risk models, and two strategies. The three expected returns are referred to as raw alpha, orthogonal alpha, and ideal alpha which is the implied alpha of the target portfolio. We define the raw alpha to be $X_A\omega$, the orthogonal alpha as $(I - P_R)X_A\omega$, and ideal alpha as Qh^* where Q is the risk model used in the portfolio construction strategy. Our two risk models are WW21AxiomaMH and WWAllAlphas. The WW21AxiomaMH model is Axioma's global fundamental equity risk model that covers the 10,000 securities that we used in our experiment. The WWAllAlphas model is a custom risk model that is constructed exactly as is WW21AxiomaMH except that the momentum, value, and growth factors in WW21AxiomaMH are replaced with the three alpha factors. Lastly, the strategies are given by (6) and (7); one is unconstrained and the other is constrained to be neutral to all risk factors given by X_R. Note that the sets of risk factors in our two risk models are equivalent.

The results for the transfer coefficients of the constrained and unconstrained cases are given in Table 1 and 2, respectively. By definition, the TC equals one for each of the unconstrained scenarios. Note also that there are no general relationships between TC and TC*. The TC may be greater than TC* in some scenarios while TC* is greater than TC in others. In our quest for obtaining an optimal portfolio that is proportional to the target, we are most interested in TC*.

In Tables 3-8, we provide other characteristics of the optimal portfolios that help explain the consistency or inconsistency of the various cases. In Tables 3 and 4, we provide the weights on the pure alpha FMPs determined by regressing each optimal portfolio against the pure alpha FMPs. The R-squared column gives the R-squared value of this regression that tells us how well the optimal portfolio can be explained by the pure alpha FMPs. The Risk Ratio column is the ratio of the predicted risk of the optimal portfolio as determined by WWAllAlphas to that determined by WW21AxiomaMH. The risk ratio is the bias statistic if we think of the WWAllAlphas risk estimate as the true risk estimate because it captures all of the systematic risk in the alpha factors.

In Tables 5 and 6, we provide exposures of the optimal portfolios to the raw alpha signals. This is interesting because exposures of the target portfolio to the raw alpha signals equal the weights on the alpha factor FMPs, ω.[11] Therefore, if an optimal portfolio is proportional to h^*, then its exposures to the raw alpha signals should be proportional to ω. In Tables 7 and 8, we give the percent of total predicted variance of the optimal portfolios that is attributable to each of the three alpha factors. Since the alpha factors are only present in the WWAllAlphas model, we use this risk model for all percent of variance calculations regardless of the risk

[11] To see this, let

$$h^* = \Delta^{-2}X(X^{\mathrm{T}}\Delta^{-2}X)^{-1}\omega.$$

Then the exposures of the target portfolio to all factors in the risk model can be written as

$$X^{\mathrm{T}}h^* = X^{\mathrm{T}}\Delta^{-2}X(X^{\mathrm{T}}\Delta^{-2}X)^{-1}\omega = \omega.$$

model used in the portfolio construction strategy.

	WWAllAlphas		WW21AxiomaMH	
Alpha	**TC**	**TC***	**TC**	**TC***
Ideal	1.00	1.00	1.00	1.00
Raw	0.77	0.93	0.96	0.58
Orthogonal	0.74	0.96	0.97	0.56

Table 1: Transfer coefficients for constrained portfolio construction

	WWAllAlphas		WW21AxiomaMH	
Alpha	**TC**	**TC***	**TC**	**TC***
Ideal	1.00	1.00	1.00	1.00
Raw	1.00	0.72	1.00	0.56
Orthogonal	1.00	0.71	1.00	0.55

Table 2: Transfer coefficients for unconstrained portfolio construction

The ideal alpha is the only alpha for which both TC and TC* are equal to one in both the constrained and unconstrained strategies and for both risk models. The reason for the TC equaling one is twofold. First, the target portfolio was constructed to be neutral to the risk factors. Second, setting the expected returns to the implied alpha of the target portfolio ensures that the optimal solution is h^*. TC* equaling one then follows by construction. TC and TC* do not tell the whole story, however. The exposures and FMP weights are proportional to the case that used WWAllAlphas and the contributions to risk are equivalent between the two cases as we expected. However, when using the WW21AxiomaMH risk model and the constrained strategy, the Risk Ratio is 1.32 which means that WW21AxiomaMH significantly underestimates the risk of the optimal portfolio.

The use of either the raw alpha or the orthogonal alpha in the constrained strategy with the WWAllAlphas risk model is interesting because the portfolio R-squared values are equal to one. This means that the optimal portfolio can be written as a linear combination of the pure alpha signals. However, note that the regression weights, exposures to raw alpha signals, and contributions to risk are quite different from the case where the ideal alpha is used with the same strategy and risk model. More weight is given to the ROE signal and less to the sales-to-price and momentum signals when using the orthogonal alpha. The cause of this is the weights applied to the columns of $\hat{X}_A$ in constructing the orthogonal alpha. If we apply the weights γ given by (19) instead of ω to the alpha signals, we recover h^* in the constrained strategy. However, the transfer coefficient will be less than one in this case. In fact, in the case that used the WWAllAlphas risk model, the transfer coefficient was only 0.747.

The raw alpha is perhaps the most frequently used alpha of the three. As we previously discussed, the unconstrained optimum will generally be exposed to the risk factors. Therefore, when we constrain the portfolio to be risk factor neutral, the TC is reduced from one. In our constrained scenario, the TC is larger when using WW21AxiomaMH than when using WWAllAlphas.[12] Despite this, the TC* is much greater when using WWAllAlphas than when using WW21AxiomaMH. The reason is due to factor alignment. Because the systematic risk in the alpha is accounted for when using WWAllAlphas, the optimizer has little incentive to bet on anything other than the pure alpha signal. When using a risk model that is not aligned, the optimizer has incentive to bet on the portion of the raw alpha that is orthogonal to the factors in the risk model. This leads to a portfolio that represents the orthogonal portion of alpha rather than the entire alpha (see Saxena and Stubbs, 2013). This is supported by the other portfolio statistics as well. First, the risk model did not capture the risk accurately resulting in a risk ratio of 2.24. Unlike the ideal alpha scenario, the weights on the pure alpha signals were not proportional to the target portfolio. The portfolio exposures and contributions to variance are very different as well. Both statistics indicate a significant overweight of sales-to-price and underweight of ROE and momentum. A closer proximity to the target portfolio when using an aligned risk model is not accidental. We find that the TC* being greater when using an aligned model generally holds in more realistic cases as well.

Alpha	Risk Model	Pure Alpha Portfolio Weights			Portfolio R-Squared	Risk Ratio
		ROE	Sales-to-Price	Momentum		
	h^*	0.69	0.22	0.09	1.00	N/A
Ideal	WWAllAlphas	4.64	1.51	0.59	0.99	1.00
Raw	WWAllAlphas	5.36	0.73	0.06	1.00	1.00
Ortho	WWAllAlphas	5.17	1.11	0.09	1.00	1.00
Ideal	WW21AxiomaMH	6.13	2.00	0.77	0.99	1.32
Raw	WW21AxiomaMH	11.65	2.86	-0.15	0.78	2.24
Ortho	WW21AxiomaMH	10.32	5.44	-0.19	0.82	2.61

Table 3: Results of regressing portfolios weights against FMP weights for constrained strategies

Alpha	Risk Model	Pure Alpha Portfolio Weights			Portfolio R-Squared	Risk Ratio
		ROE	Sales-to-Price	Momentum		
Ideal	WWAllAlphas	4.64	1.51	0.59	0.99	1.00
Raw	WWAllAlphas	7.14	0.49	0.63	0.86	1.00
Ortho	WWAllAlphas	6.98	1.32	0.84	0.83	1.00
Ideal	WW21AxiomaMH	6.13	2.00	0.77	0.99	1.32
Raw	WW21AxiomaMH	11.86	2.75	0.01	0.83	2.20
Ortho	WW21AxiomaMH	10.70	5.28	0.00	0.87	2.60

Table 4: Results of regressing portfolios weights against FMP weights for unconstrained strategies

[12] The TC being less when using an aligned model is dependent on the strategy and is not generally observed in realistic strategies that are more heavily constrained.

Alpha	**Risk Model**	**Exposures to Alpha Factors**		
		ROE	**Sales-to-Price**	**Momentum**
	h^*	0.69	0.22	0.09
Ideal	WWAllAlphas	4.64	1.51	0.59
Raw	WWAllAlphas	5.36	0.73	0.06
Ortho	WWAllAlphas	5.17	1.11	0.09
Ideal	WW21AxiomaMH	6.10	1.98	0.77
Raw	WW21AxiomaMH	10.83	3.02	0.11
Ortho	WW21AxiomaMH	9.61	5.57	0.04

Table 5: Exposures to raw alpha signals of optimal portfolios using constrained strategy.

Alpha	**Risk Model**	**Exposures to Alpha Factors**		
		ROE	**Sales-to-Price**	**Momentum**
Ideal	WWAllAlphas	4.64	1.51	0.59
Raw	WWAllAlphas	6.98	0.72	0.43
Ortho	WWAllAlphas	6.43	1.13	0.48
Ideal	WW21AxiomaMH	6.10	1.98	0.77
Raw	WW21AxiomaMH	11.28	3.04	0.21
Ortho	WW21AxiomaMH	9.97	5.39	0.10

Table 6: Exposures to raw alpha signals of optimal portfolios using unconstrained strategy.

Alpha	**Risk Model**	**Variance Contribution of Alpha Factors**		
		ROE	**Sales-to-Price**	**Momentum**
	h^*	59.7	19.1	7.7
Ideal	WWAllAlphas	59.4	19.0	7.7
Raw	WWAllAlphas	80.5	3.7	0.1
Ortho	WWAllAlphas	73.5	11.2	0.1
Ideal	WW21AxiomaMH	59.4	19.0	7.7
Raw	WW21AxiomaMH	63.0	18.3	-0.1
Ortho	WW21AxiomaMH	33.2	53.0	-0.1

Table 7: Risk constributions of alpha signals in constrained strategy as measured by the WWAllAlphas risk model.

Alpha	Risk Model	Variance Contribution of Alpha Factors		
		ROE	Sales-to-Price	Momentum
Ideal	WWAllAlphas	59.4	19.1	7.7
Raw	WWAllAlphas	70.6	2.7	0.8
Ortho	WWAllAlphas	57.9	7.9	0.3
Ideal	WW21AxiomaMH	59.4	19.0	7.7
Raw	WW21AxiomaMH	63.5	20.8	-0.3
Ortho	WW21AxiomaMH	32.1	53.5	-0.4

Table 8: Risk constributions of alpha signals in unconstrained strategy as measured by the WWAllAlphas risk model.

Conclusions

We have outlined a consistent investment process that optimally transfers alpha signals into a final portfolio. We showed how deviations from a consistent investment process can lead to a significant loss of information, regardless of the value of the transfer coefficient. The transfer coefficient, as defined by Clarke et al. (2006), does not entirely capture this loss in the transfer of information; it only considers the effect of constraints on that transfer. We introduced an alternative transfer coefficient measure, TC^*, that also considers the effect of the risk model used in the portfolio construction strategy and the alpha construction process itself. We showed how small differences in risk models or alpha construction processes can dramatically impact our ability to create an optimized portfolio that is close to the target portfolio as measured by a variety of analytics.

Not following a consistent process is frequently the cause of portfolio managers having underperforming portfolios despite having information in their alpha signals. This is often caused by an inconsistent process leading to undesirable exposures. Even in the case where these exposures are controlled via constraints, the inconsistency between expected returns and risk models can lead to an allocation of the risk budget that is significantly different from that represented in the expected returns or target portfolio.

A consistent investment process leads to an efficient transfer of information from the alpha signals to the final portfolio. In order to realize the potential of a consistent investment process, the expected returns may need to be modified in order to be consistent with the risk model and portfolio construction strategy. The expected returns should be, after all, an integral part of a consistent investment process. Implementing a consistent expected returns model should not be viewed as any greater alteration of an investment process than a change in the portfolio construction strategy or risk model. In fact, as we have shown, all three components are intimately connected in the construction of optimized portfolios.

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Technical Appendix

Proposition 1. *The solution of the FMP construction problem given by* (1) *is*

$$h^* = (I - P_R)X_A[X_A^{\mathrm{T}}(I - P_R)X_A]^{-1}.$$

Furthermore, if we use a general covariance matrix Q and assume we have multiple alpha signals, then the FMP for the jth signal is

$$h^* = Q^{-1/2}[I - P_R^Q]Q^{-1/2}X_A[X_A^{\mathrm{T}}Q^{-1/2}(I - P_R^Q)Q^{-1/2}X_A]^{-1}\tilde{e}_j,$$

where $\tilde{e}_j$ is the jth column of the identity matrix whose dimension equals the number of alpha signals, i.e., the number of columns in X_A.

Proof. In the multiple alpha signal case, the constraint $X_A^{\mathrm{T}}h = 1$ becomes $X_A^{\mathrm{T}}h = \tilde{e}_j$. If we write both the risk factor and alpha factor exposure constraints in (1) as the single set of constraints $X^{\mathrm{T}}h = e_j$, we get

$$h = Q^{-1}X\left(X^{\mathrm{T}}Q^{-1}X\right)^{-1}e_j.$$

Now, we will write this out in terms of risk and alpha factors separately. Let

$$P_R^Q = Q^{-1/2}X_R(X_R^{\mathrm{T}}Q^{-1}X_R)^{-1}X_R^{\mathrm{T}}Q^{-1/2}.$$

Then, we have

$$\begin{aligned}
h^* &= -Q^{-1}X_R(X_R^{\mathrm{T}}Q^{-1}X_R)^{-1}X_R^{\mathrm{T}}Q^{-1}X_A[X_A^{\mathrm{T}}Q^{-1/2}(I - P_R^Q)Q^{-1/2}X_A]^{-1}\tilde{e}_j \\
&\qquad + Q^{-1}X_A[X_A^{\mathrm{T}}Q^{-1/2}(I - P_R^Q)Q^{-1/2}X_A]^{-1}\tilde{e}_j \\
&= [Q^{-1} - Q^{-1}X_R(X_R^{\mathrm{T}}Q^{-1}X_R)^{-1}X_R^{\mathrm{T}}Q^{-1}]X_A[X_A^{\mathrm{T}}Q^{-1/2}(I - P_R^Q)Q^{-1/2}X_A]^{-1}\tilde{e}_j \\
&= Q^{-1/2}[I - P_R^Q]Q^{-1/2}X_A[X_A^{\mathrm{T}}Q^{-1/2}(I - P_R^Q)Q^{-1/2}X_A]^{-1}\tilde{e}_j.
\end{aligned}$$

Then completes the general case. For the single alpha signal case that uses $Q = \sigma^2 I$, substituting this value of Q and $\tilde{e}_j = 1$ into the general result gives

$$h^* = (I - P_R)X_A[X_A^{\mathrm{T}}(I - P_R)X_A]^{-1}.$$

□

Proposition 2. *The residuals of the least-squares solution of* (4) *are a scalar multiple of the solution of* (1) *when Q is a multiple of I. In the more general case, where we use Δ^2 in the objective of* (1)*, the same results holds for an equivalent weighted least-squares solution of* (4)*.*

Proof. The least-squares solution is $\lambda = (X_R^{\mathrm{T}}X_R)^{-1}X_R^{\mathrm{T}}X_A$. Therefore,

$$\begin{aligned}
\varepsilon &= X_A - X_R\lambda \\
&= X_A - X_R(X_R^{\mathrm{T}}X_R)^{-1}X_R^{\mathrm{T}}X_A \\
&= (I - P_R)X_A.
\end{aligned}$$

It is then straightforward to show that $(I - P_R)X_A$ scaled by $[X_A^{\mathrm{T}}(I - P_R)X_A]^{-1}$ has unit exposure to X_A. □

Proposition 3. *If $\alpha = Qh^*$, then the solution of the unconstrained and constrained portfolio construction problems given by* (6) *and* (7) *is a multiple of h^*.*

Proof. To see the result, we will plug in the value Qh^* for α in the optimal solutions of (6) and (7). First, we consider the unconstrained case. Here,

$$\begin{aligned} h &= \frac{1}{2\lambda} Q^{-1}\alpha \\ &= \frac{1}{2\lambda} Q^{-1} Q h^* \\ &= \frac{1}{2\lambda} h^*. \end{aligned}$$

So, the optimal solution of (6) is a multiple of h^*.

In the constrained case, we can write the optimal solution as

$$\begin{aligned} h &= \frac{1}{2\lambda} Q^{-1/2} \left(I - Q^{-1/2} X_R (X_R^{\mathrm{T}} Q^{-1} X_R)^{-1} X_R^{\mathrm{T}} Q^{-1/2} \right) Q^{-1/2} \alpha \\ &= \frac{1}{2\lambda} Q^{-1/2} \left(I - Q^{-1/2} X_R (X_R^{\mathrm{T}} Q^{-1} X_R)^{-1} X_R^{\mathrm{T}} Q^{-1/2} \right) Q^{-1/2} Q h^* \\ &= \frac{1}{2\lambda} h^* - \frac{1}{2\lambda} Q^{-1} X_R (X_R^{\mathrm{T}} Q^{-1} X_R)^{-1} X_R^{\mathrm{T}} h^* \\ &= \frac{1}{2\lambda} h^*. \end{aligned}$$

The last equality uses the property that the target portfolio is not exposed to the risk factors by construction, i.e., $X_R^{\mathrm{T}} h^* = 0$. □

Proposition 4. *Given a set of FMP weights, ω, the target portfolio is the optimal solution of* (13)*.*

Proof. Let $X = [\; X_R \;\; X_A \;]$ and $\tilde{\omega}^{\mathrm{T}} = [\; 0 \;\; \omega \;]^{\mathrm{T}}$. Consider the compact representation of the FMP for alpha signal j:

$$h_j = \Delta^{-2} X \left(X^{\mathrm{T}} \Delta^{-2} X \right)^{-1} e_j. \tag{24}$$

Then

$$\begin{aligned} h^* &= \sum_j h_j \omega_j \\ &= \sum_j \Delta^{-2} X \left(X^{\mathrm{T}} \Delta^{-2} X \right)^{-1} e_j \omega_j \\ &= \Delta^{-2} X \left(X^{\mathrm{T}} \Delta^{-2} X \right)^{-1} \sum_j e_j \omega_j \\ &= \Delta^{-2} X \left(X^{\mathrm{T}} \Delta^{-2} X \right)^{-1} \tilde{\omega}. \end{aligned}$$

It is then easy to see that h^* is the optimal solution of

$$\begin{array}{ll} \text{minimize} & h^{\mathrm{T}} Q h \\ \text{subject to} & X^{\mathrm{T}} h = \tilde{\omega}. \end{array} \tag{25}$$

which is equivalent to (13). □

Proposition 5. *The optimal target portfolio is given by* (11)*.*

Proof. We have already shown that the FMP for jth alpha signal can be written as

$$h_j = (I - P_R)X_A \left[X_A^{\mathrm{T}}(I - P_R)X_A\right]^{-1} e_j.$$

Therefore, we have

$$\begin{aligned} h^* &= \sum_j h_j \omega_j \\ &= \sum_j (I - P_R)X_A \left[X_A^{\mathrm{T}}(I - P_R)X_A\right]^{-1} e_j \omega_j \\ &= (I - P_R)X_A \left[X_A^{\mathrm{T}}(I - P_R)X_A\right]^{-1} \sum_j e_j \omega_j \\ &= (I - P_R)X_A \left[X_A^{\mathrm{T}}(I - P_R)X_A\right]^{-1} \omega. \end{aligned}$$

□

Proposition 6. *Let $\hat{X}_A = (I - P_R)X_A$ where X_A is a matrix whose columns are the raw alpha signals. Then, if we let $\alpha = \hat{X}_A\gamma$ for some vector or raw alpha signal weights γ, then the optimal solution of* (7) *is proportional to h^*.*

Proof. We can write the optimal solution of (7) as

$$\begin{aligned} h &= \frac{1}{2\lambda} Q^{-1/2} \left(I - Q^{-1/2}X_R(X_R^{\mathrm{T}}Q^{-1}X_R)^{-1}X_R^{\mathrm{T}}Q^{-1/2}\right) Q^{-1/2}\hat{X}_A\gamma \\ &= \frac{1}{2\lambda} Q^{-1/2} \left(I - Q^{-1/2}X_R(X_R^{\mathrm{T}}Q^{-1}X_R)^{-1}X_R^{\mathrm{T}}Q^{-1/2}\right) Q^{-1/2}(I - X_R(X_R^{\mathrm{T}}X_R)^{-1}X_R^{\mathrm{T}})X_A\gamma \\ &= \frac{1}{2\lambda} Q^{-1/2} \left(I - Q^{-1/2}X_R(X_R^{\mathrm{T}}Q^{-1}X_R)^{-1}X_R^{\mathrm{T}}Q^{-1/2}\right) Q^{-1/2}X_A\gamma \\ &\quad - \frac{1}{2\lambda} Q^{-1/2} \left(Q^{-1/2}X_R(X_R^{\mathrm{T}}X_R)^{-1}X_R^{\mathrm{T}}X_A\gamma - Q^{-1/2}X_R(X_R^{\mathrm{T}}Q^{-1}X_R)^{-1}X_R^{\mathrm{T}}Q^{-1}X_R(X_R^{\mathrm{T}}X_R)^{-1}X_R^{\mathrm{T}}X_A\gamma\right) \\ &= \frac{1}{2\lambda} Q^{-1/2} \left(I - Q^{-1/2}X_R(X_R^{\mathrm{T}}Q^{-1}X_R)^{-1}X_R^{\mathrm{T}}Q^{-1/2}\right) Q^{-1/2}X_A\gamma. \end{aligned} \tag{26}$$

So, if we let

$$\gamma = \left[X_A^{\mathrm{T}}Q^{-1/2}(I - P_R^Q)Q^{-1/2}X_A\right]^{-1} \omega, \tag{27}$$

then the solution to (7) is proportional to h^* when $\alpha = \hat{X}_A\gamma$. □

Proposition 7. *If we let $\alpha = \hat{X}_A\gamma = X_A\gamma$ where*

$$\gamma = \left[X_A^{\mathrm{T}}Q^{-1/2}(I - P_R^Q)Q^{-1/2}X_A\right]^{-1} \omega,$$

then γ is equal to the covariance between the target portfolio and the the FMPs for the alpha signals, i.e., $\gamma_j = h_j^{\mathrm{T}}Qh^*$.

Proof. The target portfolio and FMP of alpha factor j can be written as

$$h^* = Q^{-1/2}[I - P_R^Q]Q^{-1/2}X_A[X_A^{\mathrm{T}}Q^{-1/2}(I - P_R^Q)Q^{-1/2}X_A]^{-1}\omega \text{ and}$$
$$h_j = Q^{-1/2}[I - P_R^Q]Q^{-1/2}X_A[X_A^{\mathrm{T}}Q^{-1/2}(I - P_R^Q)Q^{-1/2}X_A]^{-1}\tilde{e}_j,$$

respectively. Then the covariance between h_j and h^* can be written as

$$\begin{aligned}
h_j^{\mathrm{T}}Qh^* &= \tilde{e}_j^{\mathrm{T}}[X_A^{\mathrm{T}}Q^{-1/2}(I - P_R^Q)Q^{-1/2}X_A]^{-1}X_A^{\mathrm{T}}Q^{-1/2}[I - P_R^Q]Q^{-1/2} \\
&\quad QQ^{-1/2}[I - P_R^Q]Q^{-1/2}X_A[X_A^{\mathrm{T}}Q^{-1/2}(I - P_R^Q)Q^{-1/2}X_A]^{-1}\omega \\
&= \tilde{e}_j^{\mathrm{T}}[X_A^{\mathrm{T}}Q^{-1/2}(I - P_R^Q)Q^{-1/2}X_A]^{-1}[X_A^{\mathrm{T}}Q^{-1/2}[I - P_R^Q]Q^{-1/2}X_A][X_A^{\mathrm{T}}Q^{-1/2}(I - P_R^Q)Q^{-1/2}X_A]^{-1}\omega \\
&= \tilde{e}_j^{\mathrm{T}}[X_A^{\mathrm{T}}Q^{-1/2}(I - P_R^Q)Q^{-1/2}X_A]^{-1}\omega \\
&= \tilde{e}_j^{\mathrm{T}}\gamma \\
&= \gamma_j.
\end{aligned}$$

□

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